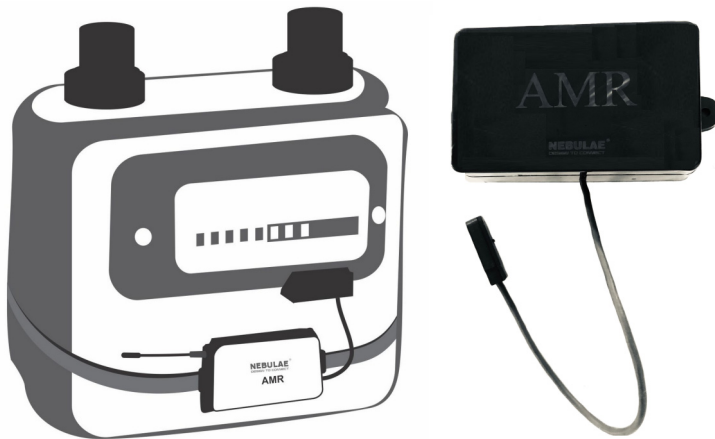


Automatic Meter Reading (AMR) Device For Gas Meter

NEBULAE®
DESIGN TO CONNECT



Nebulae® AMR is unique wireless battery based device for live reporting of Gas consumption data with built-in pulse detector which can be installed on existing diaphragm Gas meters. The device provides alerts of tamper detection on device removal and magnetic interference. It provides a long range (~10KM) using multipoint communication technology with single LoRaWAN Gateway/DCU connected with Nebulae or other service provider's LoRaWAN® server, which makes it the most cost effective solution for utility companies.

Features



Plug-n-Play and Cost Effective

- Retrofit on existing meters with easy hook option of screw/cable tie
- No configuration required, sends live data on pre-configured intervals after installation
- Wireless and battery based device



Accurate Meter Reading

- Meter reading based on incremental Pulse count from meter using Hall effect technology
- Sensor stays always on and wakes device from sleep on pulse detection
- Supported dial speed for G1.6 domestic meters with Qmax = 2.5 m3/h : 6 Seconds/revolution



Live Reporting

- Fixed or Configurable reporting interval
- Sends sum of Pulse counts calculated which results on Total Gas Consumption Count at Server
- Sends status and alerts for battery, magnetic interference and tamper detection



Tamper Proof

- Supports device removal alerts
- Sends any magnetic interference alerts
- Disables all other operations on Tamper event (configurable)



Long Battery

- Non-Rechargeable D-cell (Lithium Thionyl Chloride) Battery (Replaceable)
- 5+ years of battery life with deep sleep technology



Secure Remote Wireless Communication

- No Walk By or Drive By needed as Client – Server model
- Multipoint communication up to 10KM (line of sight) with single LoRaWAN® Gateway/DCU to reduce overall infrastructure cost
- Works with any LoRaWAN® Gateway/DCU to provide connectivity with Nebulae LoRaWAN private server
- Build on NebuLink Node NLN500a, first and smallest STM32WL based LoRaWAN Module
 - Transmit Power, up to +22 dBm
 - Frequency Band, EU 863-870, US 902-928, AU 915-928, IN865



Environment Friendly

- Industrial robust and Secure Design
- Suitable for outdoor environment
- Operational temperature -20°C to +85°C
- Ingress protection IP65 or higher



No Maintenance

- Due to Plug-n-Play nature of the device, no maintenance required



Smallest Dimensions*

- 97 x 59 x 37 mm

* Customization is supported based on requirements

For more information contact at <https://www.nebulae.io/> or sales@nebulae.io